



Time Patterns in Traffic Stop Data

The Rhode Island Traffic Stops Dataset, Part Three

Real records from the Stanford Open Policing Project

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RECAP

Picking up where part two left off

- The file is fully cleaned: 86,536 rows, 13 columns, indexed by stop_datetime
- Parts one and two compared categories and groups within a single moment in time
- This notebook adds a new dimension: how patterns change across hours, years, and durations
- Every chart ahead is built from the published statistics for the complete file



SECTION 1

Working with the DatetimeIndex

Once dates are the index, calendar attributes come for free



STEP 1

Read calendar attributes from the index

```
stops.index.month
```

```
stops.index.hour
```



No dt accessor needed

The dt accessor is for a regular column. Once dates are the index, attributes come straight off it



Any calendar attribute works this way

year, month, day, hour, and weekday are all available directly on a DatetimeIndex



Your turn: read attributes off a DatetimeIndex

Open a notebook and try each step on your own data



SECTION 2

Does time of day affect arrest rate

Grouping by `index.hour` compares a rate across all 24 hours at once



STEP 1

Group by hour to compare arrest rates

```
stops.groupby(stops.index.hour)
```

```
.is_arrested.mean()
```

This splits every stop into one of 24 buckets by the hour it happened, then takes the mean of `is_arrested` within each one, on the complete file

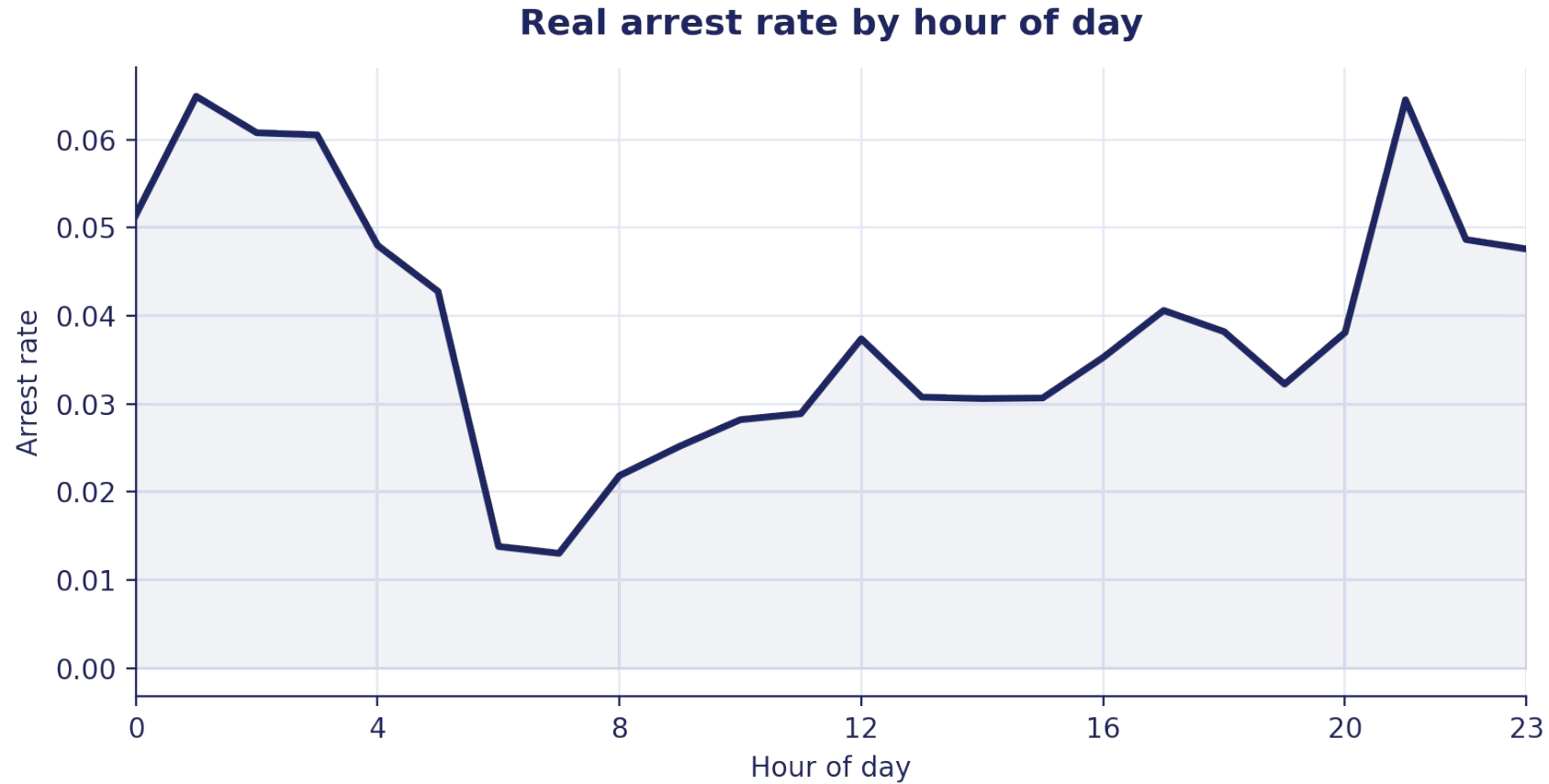
```
00  0.051431
01  0.064932
02  0.060798
03  0.060549
04  0.048000
05  0.042781
06  0.013813
07  0.013032
21  0.064541
22  0.048666
23  0.047592
```

Showing hours 0 to 7 and 21 to 23. The full table has all 24



STEP 2

Plot the hourly arrest rate



The rate spikes near 1 AM and again near 9 PM, then dips to its lowest point around 6 and 7 AM



Your turn: compare a rate by hour of day

Open a notebook and try each step on your own data



SECTION 3

Are drug related stops on the rise

resample groups a DatetimeIndex into fixed calendar periods like years



STEP 1

Resample a Boolean column by year

```
stops.drugs_related_stop  
.resample("YE").mean()
```

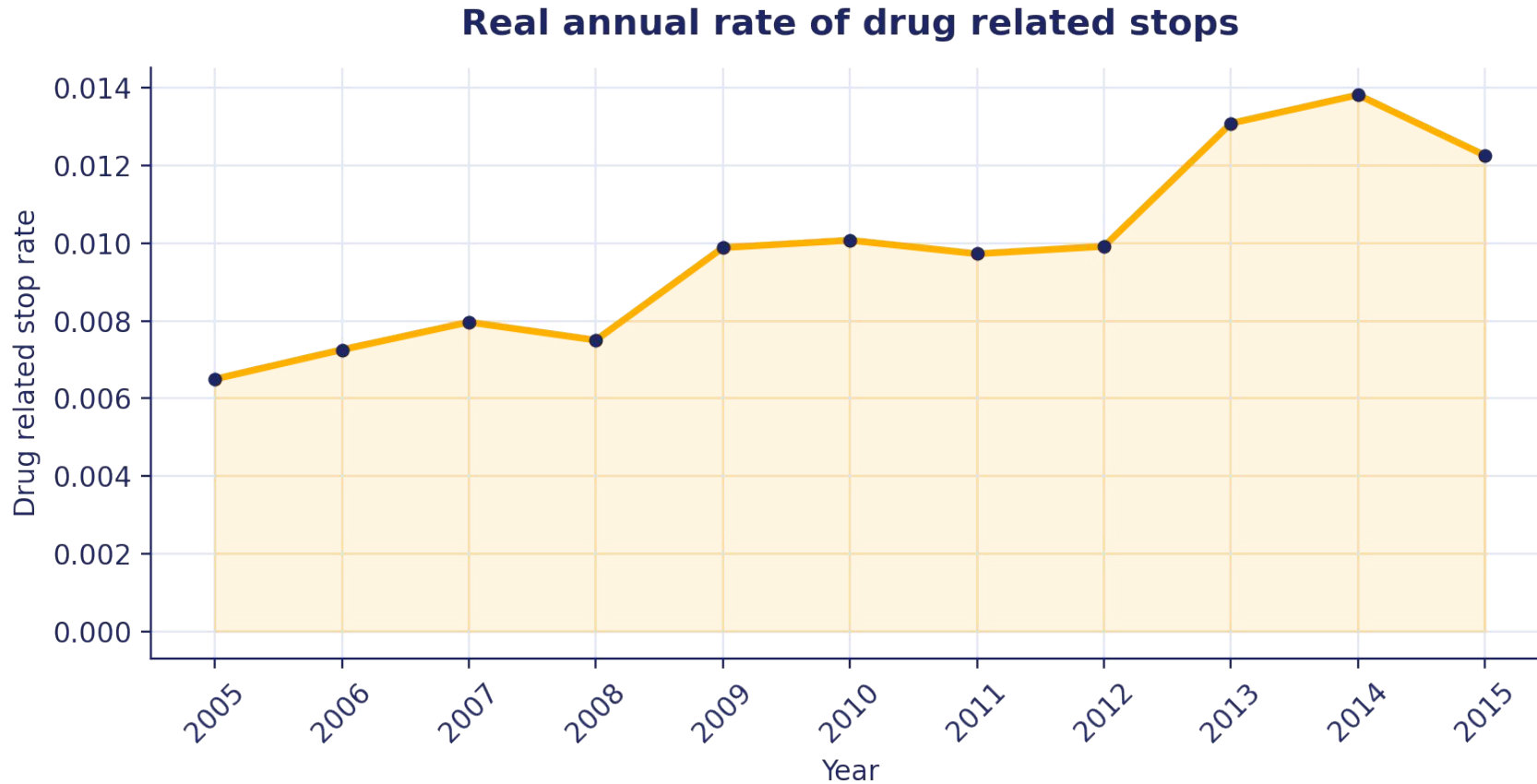
resample groups by fixed calendar periods such as years, months, or weeks, which is a natural fit once dates are the index

2005	0.006501
2006	0.007258
2007	0.007970
2008	0.007505
2009	0.009889
2010	0.010081
2011	0.009731
2012	0.009921
2013	0.013094
2014	0.013826
2015	0.012266



STEP 2

Plot the annual drug rate



The rate climbs from about 0.65 percent in 2005 to about 1.23 percent in 2015, nearly doubling



GOOD PRACTICE

Test a competing explanation before concluding

```
annual_search_rate = stops.search_conducted.resample("YE")  
                        .mean()
```



One hypothesis: more searches, more drug finds

If officers simply searched more often, drug related stops would rise too, even with no change



On the file, the search rate fell

Drug related stops climbed while the search rate moved the other way over the same eleven years

The simple explanation does not hold up. Something else is driving the increase



Your turn: resample a rate and test an explanation

Open a notebook and try each step on your own data



SECTION 4

What violations are caught in each district

crosstab builds a frequency table from two columns at once



STEP 1

Build a frequency table with crosstab

```
pd.crosstab(stops.driver_race,  
            stops.driver_gender)
```

driver_race runs down the rows and driver_gender runs across the columns. Each cell is the count of stops for that combination

driver_race	F	M
Asian	551	1838
Black	2681	9604
Hispanic	1953	7774
Other	53	212
White	18536	43334



STEP 2

Cross district with violation, then slice

```
all_zones = pd.crosstab(stops.district, stops.violation)
k_zones = all_zones.loc['Zone K1':'Zone K3']
```

district	Equipment	Moving violation	Other	Registration/plates	Seat belt	Speeding
Zone K1	672	1254	290	120	0	5960
Zone K2	2061	2962	942	768	481	10448
Zone K3	2302	2898	705	695	638	12322

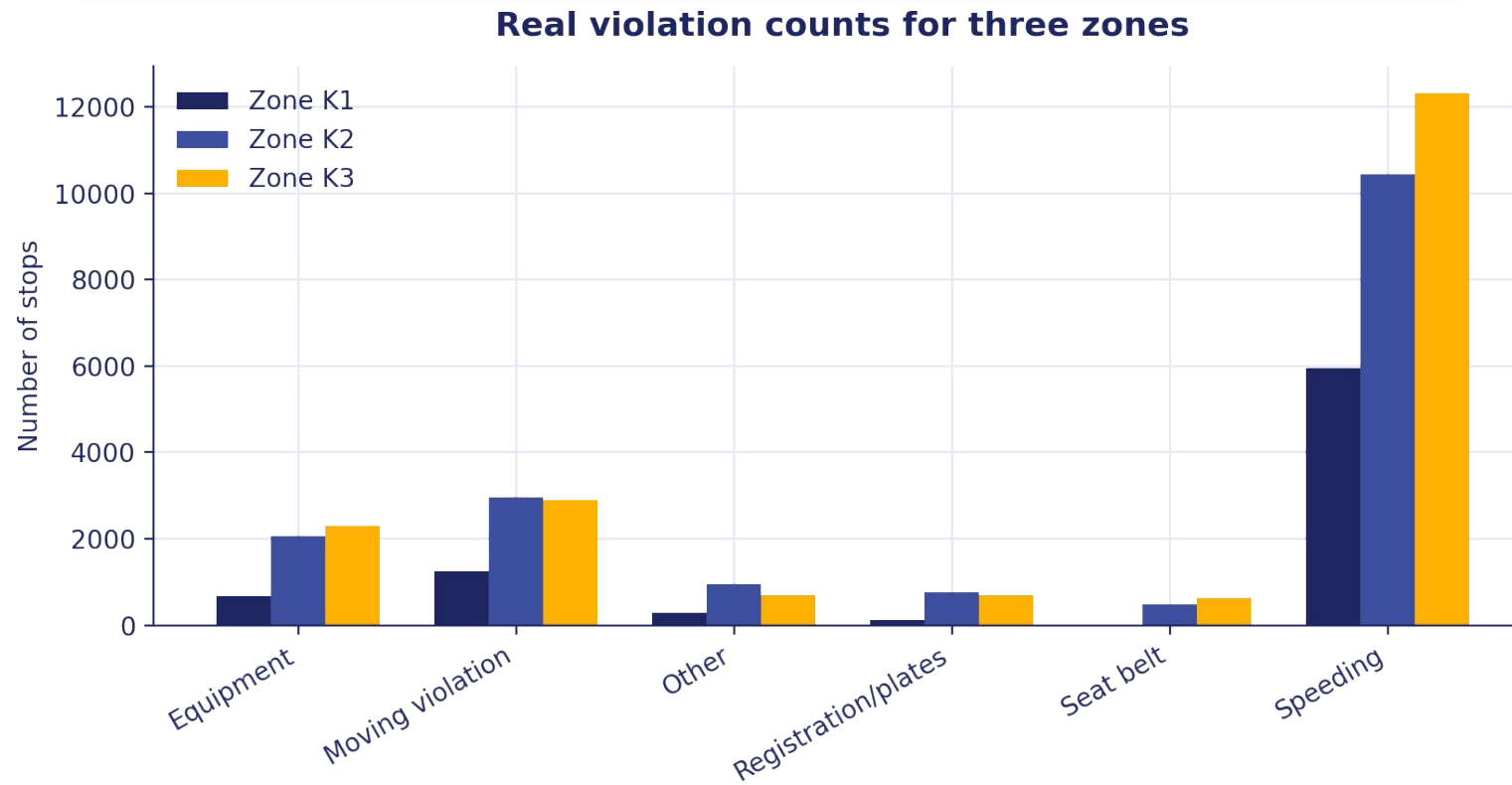
loc slices by label, keeping every row from Zone K1 through Zone K3 inclusive



STEP 3

Plot the frequency table as bars

```
k_zones.plot(kind='bar')
```

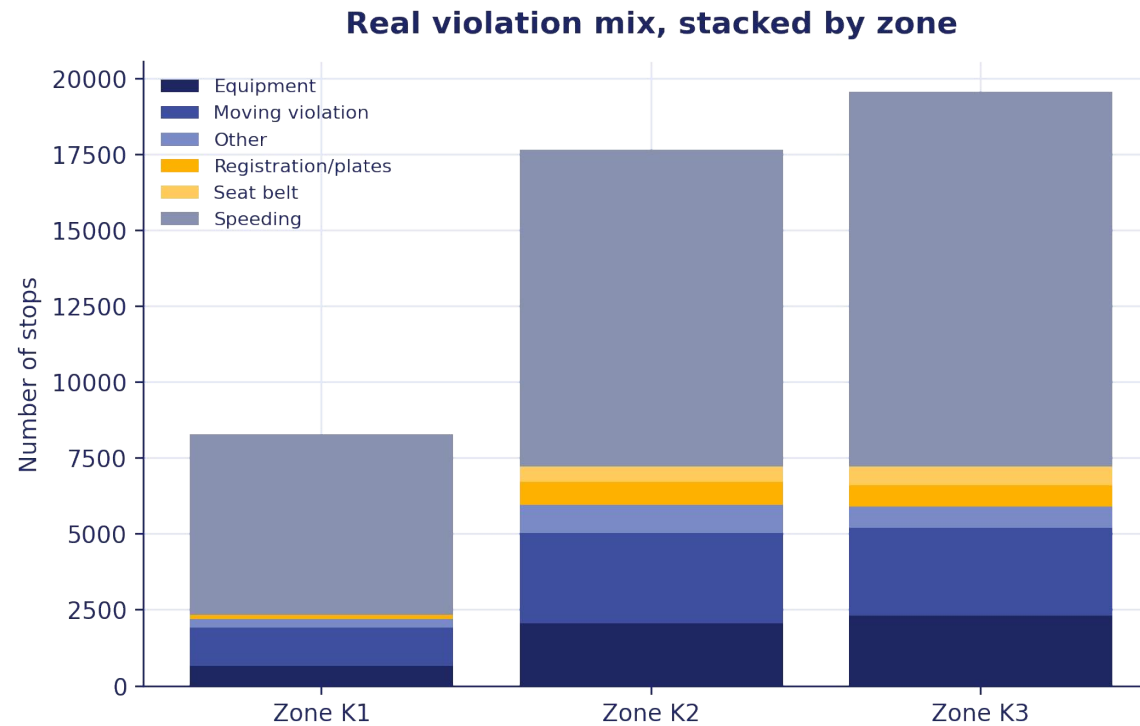




STEP 4

Stack the bars to see the total mix

```
k_zones.plot(kind='bar', stacked=True)
```



Zone K1 is overwhelmingly speeding stops. Zones K2 and K3 show a similar, more even mix



Your turn: cross two columns and plot the result

Open a notebook and try each step on your own data



SECTION 5

How long might you be stopped for a violation

Mapping turns approximate text ranges into numbers you can average



STEP 1

Map text ranges to representative numbers

```
mapping = {'Under 15 Min': 8, '15 to 30 Min': 23,  
          'Over 30 Min': 45}  
stops['stop_minutes'] = stops.stop_duration.map(mapping)
```



A dictionary maps what you have to what you want
Each text range becomes a single representative number of minutes



The exact duration is not available
The midpoint of each range is used as a reasonable estimate



STEP 2

Find the mean stop length by reason

```
stops.groupby('violation_row').stop_minutes.mean()
```

Seatbelt Violation	9.66
Speeding	10.58
Equipment/Inspection	11.45
City/Town Ordinance	13.25
Registration Violation	13.74
Other Traffic Violation	13.84

Suspicious Person	14.91
Special Detail/Patrol	15.12
Motorist Assist	17.74
APB	17.97
Call for Service	22.12
Warrant	24.06

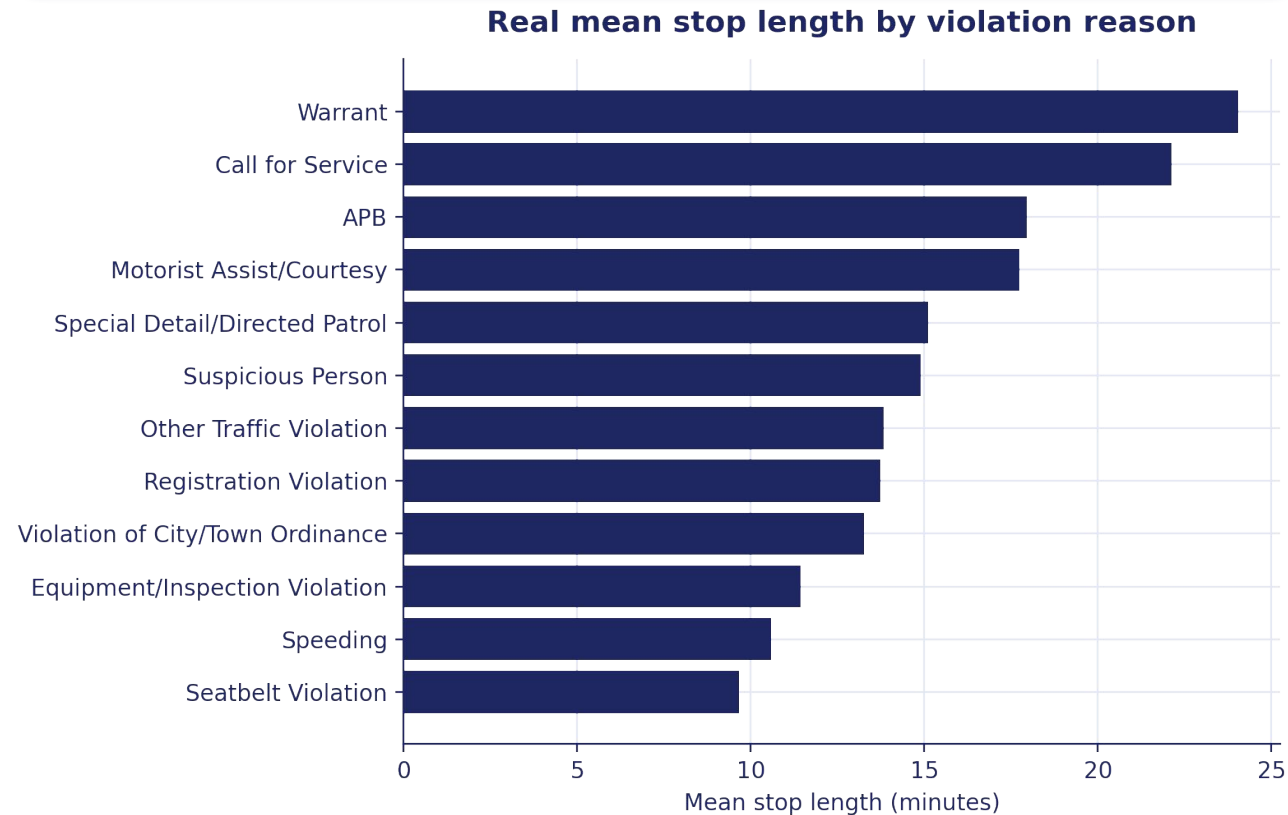
Minutes, sorted from shortest to longest average stop



STEP 3

Sort and rotate the bars for readability

```
stop_length.sort_values().plot(kind='barh')
```





Your turn: map, group, sort, and plot

Open a notebook and try each step on your own data

Five ways to bring time into the analysis



Read the index

Calendar attributes with no dt accessor



Group by hour

Compare a rate across the day



Resample by year

Track a rate over time, test explanations



Cross tabulate

Build and slice a frequency table



Map and compare

Turn text ranges into real numbers

Three notebooks, one dataset: cleaned, compared, and explored over time

Source: openpolicing.stanford.edu, Open Data Commons Attribution License